

Question 1

Marks

- (a) Evaluate, leaving answers in exact form:

(i) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin 2x \, dx$

2

(ii) $\int_1^2 \frac{e^{2x}}{e^{2x} + 1} \, dx$

2

- (b) Differentiate $y = e^x \ln x$ with respect to x .

2

- (c) Evaluate $\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(-\frac{1}{2}\right)$, leaving your answer in terms of π .

1

- (d) Use mathematical induction to show that $3^{2n+4} - 2^{2n}$ is divisible by 5 for all positive integers n .

5

Question 2 (Start a new page)

- (a) Find the second derivative of $\sin^{-1} x$.

3

- (b) Sketch the graph of $y = \frac{x-2}{x^2}$, showing any maximum and minimum turning points, points of inflexion, asymptotes, and intercepts with the coordinate axes.

6

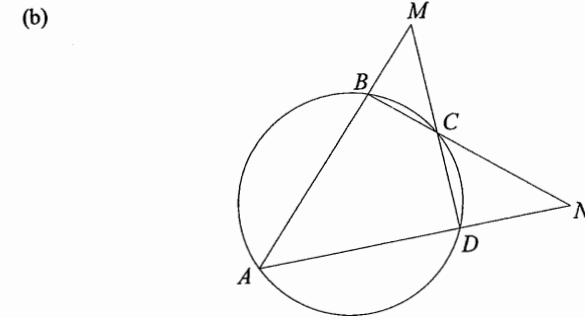
- (c) Use the substitution $u = 1 - x^2$ to find $\int \frac{x}{\sqrt{1-x^2}} \, dx$

3

Question 3 (Start a new page)

- (a) If $x^2 + y^2 = 7xy$, show that $\ln(x+y) = \ln 3 + \frac{1}{2} \ln x + \frac{1}{2} \ln y$.

3



In the figure, ABM , DCM and AND are straight lines.

Copy the diagram. Given that $\widehat{AMD} = \widehat{BNA}$, prove that

(i) $\widehat{ABC} = \widehat{ADC}$

2

- (ii) AC is the diameter of the circle.

3

- (c) Prove that $\tan^{-1} 4 - \tan^{-1} \frac{3}{5} = \frac{\pi}{4}$.

4

Question 4 (Start a new page)

- (a) The equation $\tan x - 2x = 0$ has a solution near 1.1. Use one application of Newton's method to find a better approximation to this solution.

3

- (b) For the function $y = \cos(\sin^{-1} x)$

- (i) state the domain

1

- (ii) state the range

1

- (iii) draw a neat sketch of the function

2

- (c) The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ meets the x axis at Q . Find the cartesian equation of the locus of the midpoint of PQ .

5

Question 5 (Start a new page)

- (a) Find the coefficient of x^2 in the expansion of $(1 - 2x)^{18} (1 + 3x)^{17}$. 4
- (b) Factorize the polynomial $P(x) = x^3 - x^2 - 8x + 12$ completely, given that the equation $P(x) = 0$ has a repeated root. 4
- (c) If α, β and γ are the roots of $x^3 - x^2 + 4x - 1 = 0$, find the value of
- (i) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
- (ii) $\alpha\beta\gamma$ 1
- (ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

Question 6 (Start a new page)

- (a) Prove that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. 4
- (b) Find an approximation for the area between the curve $y = 3 \sin x$ and the x axis, between $x = 0$ and $x = 1$, using the trapezoidal rule with three function values. 4
- (c) nC_k is the coefficient of x^k in the binomial expansion of $(1 + x)^n$, where n is a positive integer. By differentiating the identity $x(1 + x)^n = \sum_{k=0}^n {}^nC_k x^{k+1}$, show that $\sum_{k=0}^n (k+1) {}^nC_k = (n+2) \cdot 2^{n-1}$. 4

Question 7 (Start a new page)

- (a) An object is placed in surroundings which remain at a constant temperature of 20°C . The temperature of the object ($T^\circ\text{C}$) after t minutes is given by $T = 20 + (A - 20)e^{-kt}$, where A and k are positive constants.
- (i) Prove that $\frac{dT}{dt} = -K(T - 20)$. 1
- (ii) Initially, the temperature of the object is 50°C and is falling at a rate of 6°C per minute. Find
- (a) the values of A and K 2
- (b) the temperature (to the nearest degree) of the object after 10 minutes. 1
- (c) the time required (to the nearest minute) for the temperature of the object to reach 21°C . 2
- (b) A stone is thrown at 20 ms^{-1} at an angle of 30° above the horizontal from the edge of the top of a building 40 metres high.
- (i) Derive equations for the vertical and horizontal displacement of the stone in terms of time. Ignore air resistance, and assume that the acceleration due to gravity is 10 ms^{-2} . 2
- (ii) How long after projection will the stone strike the ground? 2
- (iii) Find the horizontal range of the flight. 2

$$\begin{aligned} \frac{1}{\pi} \int_{\pi/3}^{\pi/2} \sin 2x \, dx \\ = \left[-\frac{1}{2} \cos 2x \right]_{\pi/3}^{\pi/2} \\ = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \\ = \frac{1}{4} \end{aligned}$$

d) Let $\frac{3^{2n+4} - 2^{2n}}{5} = M \quad M \in \mathbb{Z}$

$$3^{2n+4} - 2^{2n} = 5M$$

Step 1. Prove true for $n=1$

$$3^6 - 2^2 = \frac{725}{5}$$

True for $n=1$

Assume true for $n=k$

$$3^{2k+4} - 2^{2k} = 5M$$

Prove true for $n=k+1$

$$\begin{aligned} 3^{2k+6} - 2^{2k+2} &= 9 \cdot 3^{2k+4} - 4 \cdot 2^{2k} \\ &= 9(5M + 2^{2k}) - 4 \cdot 2^{2k} \\ &= 45M + 5 \cdot 2^{2k} \\ &= 5(9M + 2^{2k}) \end{aligned}$$

which is $\div$ by 5 true for $n=k+1$

Step 3. As the result is true for $n=1$ and we assumed the result for $n=k$ and proved it for the next value $n=k+1$, then it is true for $n=2$ and so on.

$$\frac{\pi}{3} + \frac{\pi}{6} = \frac{\pi}{2}$$

Q2)

a) $\frac{d}{dx} \sin^{-1} x$

$$\begin{aligned} &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx} \left((1-x^2)^{-1/2} \right) &= -\frac{1}{2} (1-x^2)^{-3/2} \\ &= \frac{x}{(1-x^2)^{3/2}} \end{aligned}$$

$$y = \frac{x-2}{x^2}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{x^2 \cdot 1 - (x-2) \cdot 2x}{x^4} \\ &= \frac{-x(x-4)}{x^4} \end{aligned}$$

Want $\frac{dy}{dx} = 0$

$$x-4=0$$

$$x=4 \quad y = \frac{1}{8}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{x^3(-1) + (x-4)3x^2}{x^6} \\ &= \frac{-x^3 + 3x^3 - 12x^2}{x^6} \\ &= \frac{x^2(2x-12)}{x^6} \end{aligned}$$

at $x=4$

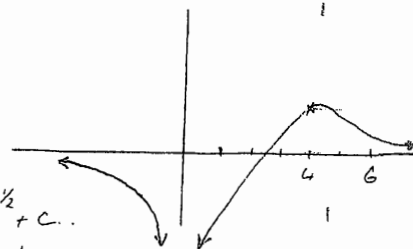
$$\frac{d^2y}{dx^2} < 0 \therefore \text{max}$$

Want $\frac{d^2y}{dx^2} = 0$

$$\therefore x=6$$

x	5	6	7
$\frac{d^2y}{dx^2}$	<0	=0	>0

$\therefore$ inf at $(6, \frac{1}{9})$



c)

$$\int \frac{x}{\sqrt{1-x^2}} \, dx$$

$$u = 1-x^2$$

$$du = -2x \, dx$$

$$-\frac{1}{2} du = x \, dx$$

$$-\frac{1}{2} \int \frac{du}{u}$$

$$= -u^{1/2} + C$$

$$= -(1-x^2)^{1/2} + C$$

1) a) $x^2 + y^2 = 7xy$
 $(x+y)^2 = 9xy$
 $\ln(x+y)^2 = \ln 9xy$
 $2 \ln(x+y) = \ln 9 + \ln x + \ln y$
 $\therefore \ln(x+y) = \ln 3 + \frac{1}{2} \ln x + \frac{1}{2} \ln y$

b) i/ diagram

ii/ In $\triangle AMD$ and $\triangle ANB$
 $\alpha)$ $\hat{AMD} = \hat{ANB}$ given
 $\hat{BAD}$ is common
 $\therefore \triangle AMD \sim \triangle ANB$

$\beta)$ $\hat{ABN} = \hat{ADM}$ ($\sim \triangle$'s)
 $ABCD$ is cyclic

opp angles of cyclic quad $= 180^\circ$

$$\therefore \hat{ABN} = \hat{ADM} = 90^\circ$$

$\therefore AC$ is a diameter

c) let $\alpha = \tan^{-1} 4$

$$\tan \alpha = 4$$

$$\beta = \tan^{-1} \frac{3}{5}$$

$$\tan \beta = \frac{3}{5}$$

$$\therefore \tan(\alpha - \beta) = \frac{4 - \frac{3}{5}}{1 + 4 \cdot \frac{3}{5}}$$

$$= \frac{\frac{17}{5}}{\frac{17}{5}}$$

$$= 1$$

$$\alpha - \beta = \frac{\pi}{4}$$

$$\therefore \tan^{-1} 4 - \tan^{-1} \frac{3}{5} = \frac{\pi}{4}$$

Q4) a) Let $f(x) = \tan x - 2x$

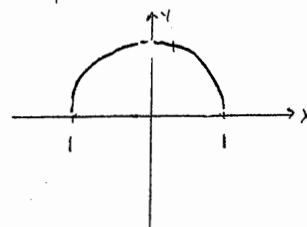
$$f'(x) = \sec^2 x - 2$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.1 - \frac{f(1.1)}{f'(1.1)}$$

$$= 1.13 \text{ to } 2\text{d}$$

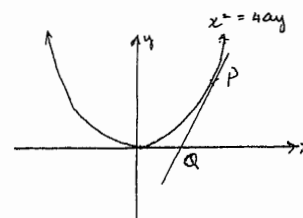
b) $y = \cos(\sin^{-1} x)$



i/ $-1 \leq x \leq 1$

ii/ $0 \leq y \leq 1$

c)



$$x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$\text{at } x = 2ap \quad \frac{dy}{dx} = p$$

$\therefore$ eqn of tangent

$$y - ap^2 = p(x - 2ap)$$

$$y = px - ap^2$$

$$\text{at } y = 0$$

$$x = ap$$

$$\therefore Q(ap, 0)$$

midpoint of PQ

$$\left(\frac{3ap}{2}, \frac{ap^2}{2} \right)$$

$\therefore$ eqn of locus

$$x = \frac{3ap}{2}$$

$$p = \frac{2x}{3a}$$

$$y = \frac{a}{2} \left(\frac{2x}{3a} \right)^2$$

$$= \frac{2x^2}{9a}$$

a)

$$2x)^{18} (1+3x)^{17}$$

$$^{18}C_1 2x + ^{18}C_2 (2x)^2 + \dots$$

$$^{17}C_1 3x + ^{17}C_2 (3x)^2 + \dots$$

$$^{17}C_2 + 4 \cdot ^{18}C_2 - 6 \cdot ^{18}C_1 \cdot ^{17}C_1$$

$$1224 + 612 - 1836$$

0

b)

$$P(x) = x^3 - x^2 - 8x + 12$$

$$P'(x) = 3x^2 - 2x - 8$$

Want $P'(x) = 0$

$$(3x+4)(x-2) = 0$$

$$x = 2 \text{ or } -\frac{4}{3}$$

$$P(2) = 8 - 4 - 16 + 12 = 0$$

$\therefore x = 2$ is the repeated root

$$P(x) = (x-2)(x-2)(x+3)$$

c)

$$x^3 - x^2 + 4x - 1 = 0$$

$$\alpha + \beta + \gamma = +1$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 4$$

$$\alpha\beta\gamma = 1$$

i/

$$\alpha^2 + \beta^2 + \gamma^2 = \frac{4}{1}$$

ii/

$$\alpha\beta\gamma = \frac{1}{1}$$

iii/

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$$

$$= 4$$

Q6) a)

$$\sin 3\theta = \sin(2\theta + \theta)$$

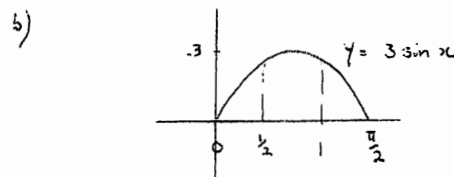
$$= \sin 2\theta \cos \theta + \sin \theta \cos 2\theta$$

$$= 2 \sin \theta \cos^2 \theta + \sin \theta (\cos^2 \theta - \sin^2 \theta)$$

$$= 2 \sin \theta \cos^2 \theta + \sin \theta \cos^2 \theta - \sin^3 \theta$$

$$= 3 \sin \theta (1 - \sin^2 \theta) - \sin^3 \theta$$

$$= 3 \sin \theta - 4 \sin^3 \theta$$



$$\int_0^{\pi/2} 3 \sin x \, dx = \frac{1}{2} \left[0 + 3 \sin 1 + 2 \times 3 \sin \frac{1}{2} \right]$$

$$= 1.350 \text{ up to } 3 \text{ D.}$$

c)

$$x(1+x)^n = {}^nC_0 x + {}^nC_1 x^2 + \dots + {}^nC_n x^{n+1}$$

by diffⁿ

$$(1+x)^n + nx(1+x)^{n-1} = {}^nC_0 + 2 {}^nC_1 x + \dots + (n+1) {}^nC_n x^n$$

$$(1+x)^{n-1} (nx + 1+x) =$$

$$(1+x)^{n-1} (1 + (n+1)x) =$$

Let $x=1$

$$2^{n-1} (n+2) = {}^nC_0 + 2 {}^nC_1 + \dots + (n+1) {}^nC_n$$

$$= \sum_{k=0}^n (k+1) {}^nC_k$$

a) i/ $T = 20 + (A-20)e^{-kt}$
 $\frac{dT}{dt} = -k(A-20)e^{-kt}$
 $= -k(T-20)$

ii/ $T = 6$ $t = 0$ $\frac{dT}{dt} = -6$ when $t = 0$

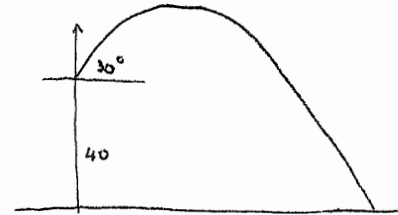
iii/ $50 = 20 + (A-20)e^0$ $-6 = -k(A-20)e^0$
 $30 = A-20$ $6 = k(A-20)$
 $A = 50$ $6 = k \cdot 30$
 $k = \frac{1}{5}$

b) $T(10) = 20 + 30e^{-\frac{1}{5} \times 10}$
 $T(10) = 20 + 30e^{-2}$
 $T = 24$

8) $21 = 20 + 30e^{-\frac{t}{5}}$
 $1 = 30e^{-\frac{t}{5}}$
 $\frac{1}{30} = e^{-\frac{t}{5}}$
 $-\ln \frac{1}{30} = \ln e^{-\frac{t}{5}}$

$t = \frac{1}{5} \ln 30$
 $= 17 \text{ min}$

b)



i/ $\ddot{x} = 0$
 $\dot{x} = t + c$
 $t = 0$ $\dot{x} = V \cos 30 = 20 \cdot \frac{\sqrt{3}}{2}$
 $\therefore c = 10\sqrt{3}$
 $x = 10\sqrt{3}t + d$
 $x = 0, t = 0 \Rightarrow d = 0$
 $x = 10\sqrt{3}t$

$\ddot{y} = -10$
 $\dot{y} = -10t + c$
 $t = 0$ $\dot{y} = 20 \times \frac{1}{2} = 10$
 $\therefore c = 10$
 $\dot{y} = -10t + 10$
 $y = -5t^2 + 10t + d$
 $t = 0$ $y = 0 \Rightarrow d = 0$
 $y = -5t^2 + 10t$

ii/ $y = -40$
 $-40 = -5t^2 + 10t$
 $t^2 - 2t - 8 = 0$
 $(t-4)(t+2) = 0$
 $\therefore t = 4 \text{ or } -2$
 but $t > 0$
 $\therefore t = 4$

iii/ $x = 10\sqrt{3}t$
 $t = 4$
 $\therefore x = 40\sqrt{3}$

6